

Sean D. Lirazan

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EDUCATION

Iowa State University | College of Engineering

Bachelor of Science in Mechanical Engineering | Minor Cyber Physical Systems

GPA: 3.71

Expected 2027

WORK EXPERIENCE

Controls Engineering Intern | *NASA Armstrong Flight Research Center*

Edwards, CA

Fall 2026

- Debugged and modified C++ sensor-acquisition code for **X-59 supersonic research aircraft**, resolving pressure-sensor data-processing and fault-handling issues and increasing usable test data by **54%** during outdoor testing.
- Supported X-56 active **flutter-suppression** research, developing a rapid-prototyping workflow using a scaled, lightweight 3D-printed aircraft model with multiple IMUs and actuated control surfaces to experimentally evaluate wing-deformation and anti-flutter control concepts.
- Adapted proprietary X-56 CAD geometry for scaled rapid prototyping, iterating physical components and electronics integration to establish a repeatable design → fabrication → test workflow for controls experimentation.
- Developed and evaluated flight-dynamics and controls simulations for Boeing aircraft applications, including autonomous taxi, takeoff, and landing systems and **737 Reduced-Gravity Flight Environments**, using MATLAB/Simulink and Python.

Systems Engineering Co-op | *Collins Aerospace (RTX)*

Cedar Rapids, IA

Spring – Fall 2026

- Conducted **mission-critical HIL simulations** and 70+ formal verification procedures for CAAS (Common Avionics Architecture System) avionics supporting 160th SOAR Night Stalkers MH-47G and MH-60M platforms
- Performed requirements-based verification of display, sensor-fusion, and mission-system software, translating software changes into laboratory tests and validating performance on representative flight hardware
- Developed proprietary automated test scripts and verification workflows, implementing **15+ process improvements** that improved test efficiency, repeatability, and laboratory throughput
- Supported controlled verification of **sensor-integrated weapons systems**, including **SAL-2 Hellfire tracking**, and avionics supporting platforms including the A/MH-6M Mission Enhanced Little Bird.

Robotics & Design Intern | *John Deere & Co.*

Waterloo, IA

Summer 2025

- Prototyped an autonomous drone for precision agriculture, health mapping across **100+** acres
- Developed a hybrid GPS-vision- based following system, implemented real time telemetry logging and storage sync with WIFI
- Integrated NDVI sensor and GPS for crop density mapping and following
- Contributed to autonomous drone navigation and AI-based camera tracking for **5+** field monitoring

Embedded Systems & Controls Specialist | *Student Innovation Lab*

ISU

Fall 2024-Present

- Served as the embedded systems expert (Arduino, STM32, AI integration) in student makerspaces
- Led **5+** embedded systems development for campus wide innovation projects, smart greenhouses and biometric tracking
- Mentored **10+** students in sensor integration (ultrasonic, imu, thermal, LiDAR), Microcontroller, programming, and RTOS

LEADERSHIP & ACTIVITIES

3DPD Club | *President*

ISU

Fall 2024-Present

- Led a 15-member engineering club though a pivot toward robotics and embedded systems, integrating 3D-printed frameworks with microcontroller-based automation
- Established industry connections by reaching out to over **10** companies, expanding learning opportunities and potential partnerships for the club.
- Spearheaded high-impact projects including a battlebot, **6-DOF robotic arm**, and an autonomous irrigation system deployed at Reiman Gardens nonprofit
- Collaborated with local charity Reiman Gardens to support local growers by offering 3D printing services.